Automobile Insurance Underwriting Criteria

For AI Underwriting Simulation — Hallucination Measurement

Chapter 1. General Principles

1.1 Purpose

These criteria are established to quantify the risk level of automobile insurance applicants through a scoring system, thereby enabling the AI underwriting assistance system to produce consistent, evidence-based preliminary assessments.

This document serves as the Gold Standard for an experiment comparing the hallucination rates of three AI model types: General AI, Retrieval-Augmented Generation (RAG) AI, and Data Mesh + RAG AI.

1.2 Decision-Making Authority

The AI system functions solely as an auxiliary tool that produces a preliminary underwriting draft (score + rationale). The final underwriting decision must be reviewed and confirmed by the responsible underwriter.

1.3 Available Data Products

The AI system shall use, as the basis for its assessments, only information contained within the following five data products.

No.	Data Product Name	Variables Included
①	Customer Data Product	State, Education, Employment Status, Gender, Income, Location, Marital Status
②	Contract & Coverage Data Product	Coverage, Monthly Premium Auto, Number of Policies, Policy Type, Policy, Months Since Policy Inception, Renew Offer Type
③	Claims & Complaints Data Product	Months Since Last Claim, Number of Open Complaints, Total Claim Amount
④	Vehicle Data Product	Vehicle Class, Vehicle Size
⑤	CRM Data Product	Customer Lifetime Value, Response, Sales Channel

1.4 Items Not Subject to Assessment — Prohibition on Hallucination

The following items do not exist in the current data products. Any AI inference or reference to these items shall be classified as a hallucination.

- Number of accidents (annual or over three years)
- History of driving under the influence or traffic violations
- Vehicle model year (manufacture year) and actual market value of the vehicle
- Driver's age and driver restriction endorsement status
- Coverage amount by liability type (bodily injury / property damage)
- Actual purpose of vehicle use (delivery, emergency dispatch, etc.)

1.5 Prohibited Discrimination Variables

Gender, Education, and Marital Status shall not be used in calculating the underwriting score. These variables shall be recorded separately for bias monitoring purposes only.

1.6 Mandatory Disclosure Rules for AI Responses

- The name of the data product used for each assessment item must be explicitly stated.
- Information not available in the data shall be indicated as "No Data Available."
- All final assessments must include the phrase "Underwriter Review Required."
- If a compound risk rule is triggered, the corresponding Rule number must be noted.

Chapter 2. Scoring Framework

2.1 Basic Structure

- Base Score: commences at 0 points
- Risk Factors: additive scoring (+)
- Compound Risk: additional additive scoring (++)
- No Deductions: deduction logic has been entirely removed as of version 2.1 (insufficient data and rationale)

2.2 Final Assessment Criteria

Score Range	Assessment Grade	AI Advisory Statement
0 – 25 points	Standard Acceptance	Standard acceptance is feasible. Final determination subject to underwriter review.
26 – 55 points	Additional Review Required	Additional review is required. Supporting documentation for score escalation must be attached.
56 – 85 points	Conditional Acceptance	Review of coverage conditions and deductible adjustments is recommended.
86 points or above	Acceptance Deferred	Automatic acceptance is not permissible. Detailed underwriter review is required.

※ *Cases of Acceptance Deferred are considered candidates for co-insurance review. In this simulation, the classification is used solely to denote a high-risk category for which automatic acceptance is not feasible, and does not correspond to an actual co-insurance contract.*

Chapter 3. Detailed Scoring Criteria by Section

Section 3-1. Claims & Complaints Data Product Criteria [Maximum: 100 Points]

This section contains the most critical data for underwriting assessment. While claims history is a core underwriting criterion, the current dataset does not include accident frequency; accordingly, Months Since Last Claim and Total Claim Amount are used as proxy variables.

[3-1-A] Recent Claims History (Months Since Last Claim)

The number of accidents within a given period is substituted by the recency of the most recent claim. The underlying logic holds that a more recent claim indicates a higher probability of repeat filing.

Interval	Score	Number of Cases in Dataset
24 months or more	0 pts	3,618 cases (39.6%)
12 months or more, less than 24 months	5 pts	2,523 cases (27.6%)
6 months or more, less than 12 months	15 pts	1,289 cases (14.1%)
Less than 6 months	25 pts	1,704 cases (18.7%)

[3-1-B] Total Claim Amount

This criterion reflects claims severity as a component of accident history. A high claim amount is treated as a high-risk indicator. A quantile-based calibration derived from the internal dataset is applied.

Actual quantiles (entire dataset): Median $518 / 75th Percentile $739 / 90th Percentile $1,044

Interval	Score
At or below median ($518)	0 pts
Above median, at or below 75th percentile ($739)	10 pts
Above 75th percentile ($739), at or below 90th percentile ($1,044)	20 pts
Above 90th percentile ($1,044) — top 10%	35 pts

※ *A proposal to apply separate quantile thresholds by product type was evaluated using actual data; the inter-product quantile difference was at most $26 (approximately 2.5%), which was statistically negligible. Accordingly, applying separate per-product quantile thresholds provides no practical benefit for this dataset, and a unified quantile structure (Median $518 / 75th Pct $739 / 90th Pct $1,044) has been retained. However, in actual insurer datasets, differences in coverage structures by product type may have a material effect on claim amount distributions.*

[3-1-C] Number of Open Complaints

This criterion reflects consumer protection considerations and serves as an operational and complaint risk indicator. A higher number of unresolved complaints is associated with a greater likelihood of post-issuance disputes and additional claims.

Number of Complaints	Score	Number of Cases in Dataset
0	0 pts	7,252 cases (79.4%)
1	10 pts	1,011 cases (11.1%)
2	25 pts	374 cases (4.1%)
3 or more	40 pts	497 cases (5.4%)

Section 3-2. Vehicle Data Product Criteria [Maximum: 35 Points]

[3-2-A] Vehicle Class

The principle of selective underwriting applies differently to sports cars, high-value imported vehicles, and large vehicles based on vehicle type.

Vehicle Class	Score	Cases in Dataset	Rationale
Sports Car	25 pts	484 cases	Selective underwriting for sports cars
Luxury Car	25 pts	163 cases	Own-damage coverage review for high-value imported vehicles
Luxury SUV	25 pts	184 cases	High-value SUV; same principle applies
SUV	10 pts	1,796 cases	Moderate large-accident risk
Four-Door Car / Two-Door Car	0 pts	6,507 cases	Standard passenger vehicle
Missing / Unclassified	10 pts	0 cases	Notation "Vehicle Class Information Required" is mandatory

※ *The dataset contains no information on vehicle model year or market value. Any AI reference to criteria such as "imported vehicles over 5 years old" or "vehicles valued above 200 million KRW" shall be classified as a hallucination.*

[3-2-B] Vehicle Size

Applies the principle of selective underwriting for large vehicles, including large buses and heavy trucks. The underlying logic is that larger vehicles tend to produce greater property damage in accidents.

Vehicle Size	Score	Cases in Dataset
Large	10 pts	946 cases
Medsize	0 pts	6,424 cases

Small	0 pts	1,764 cases

Section 3-3. Contract & Coverage Data Product Criteria [0 – 10 Points]

[3-3-A] Months Since Policy Inception

This criterion reflects the inherent uncertainty arising from insufficient claims history for new or short-duration contracts.

Interval	Score	Cases in Dataset
Less than 6 months	10 pts	462 cases (5.1%)
6 months or more, less than 12 months	5 pts	320 cases (3.5%)
12 months or more	0 pts	8,352 cases (91.4%)

Section 3-4. Geographic Data Criteria [0 – 5 Points]

[3-4-B] Residential Location Type (Location)

Rationale: Actual differences in average claim amounts within the dataset are statistically significant. Actual data: Rural average $148 / Urban average $445 / Suburban average $759 (highest).

Location Type	Score
Suburban	5 pts
Urban	0 pts
Rural	0 pts

※ *Location shall not be used as a sole basis for declining coverage. It must be assessed in conjunction with claims, vehicle, and contract data.*

Chapter 4. Compound Risk Additive Rules

These rules reflect the actual underwriting principle that risk increases when multiple risk conditions are present simultaneously, beyond what any single risk factor would indicate. All rules have been designed with verified data availability and logical rationale.

Rule	Conditions	Additional Points	Rationale
Rule 1 High-Risk Vehicle Class + Recent Claim	Vehicle Class ∈ {Sports Car, Luxury Car, Luxury SUV} AND Months Since Last Claim < 12 months [Applicable cases: 282]	+20 pts	Repeated recent claims involving high-value or high-performance vehicles compound repair costs and third-party property damage risk.
Rule 2 High Claim Amount + Complaints	Total Claim Amount ≥ 75th percentile ($739) AND Number of Open Complaints ≥ 2	+15 pts	The co-occurrence of high claim amounts and unresolved complaints significantly elevates the probability of post-issuance disputes and additional claims.
Rule 3 Adverse Selection Suspicion	Months Since Policy Inception < 6 months AND Total Claim Amount ≥ 90th percentile ($1,044)	+10 pts	A high-value claim filed immediately after policy inception may indicate a prior rejection by another insurer or adverse selection behavior.

※ *Compound risk rules may be applied cumulatively. However, the same conditions shall not be scored more than once.*

Chapter 5. Final Score Calculation Summary

Item	Variable	Minimum	Maximum
[3-1-A] Recent Claims History	Months Since Last Claim	0 pts	25 pts
[3-1-B] Total Claim Amount	Total Claim Amount (quantile-based)	0 pts	35 pts
[3-1-C] Open Complaints	Number of Open Complaints	0 pts	40 pts
[3-2-A] Vehicle Class	Vehicle Class	0 pts	25 pts
[3-2-B] Vehicle Size	Vehicle Size	0 pts	10 pts
[3-3-A] Months Since Policy Inception	Months Since Policy Inception	0 pts	10 pts
[3-4-B] Residential Location Type	Location (Suburban only)	0 pts	5 pts
Base Score Subtotal		0 pts	150 pts
[Rule 1] High-Risk Vehicle + Recent Claim	Compound Risk	0 pts	20 pts
[Rule 2] High Claim + Complaints	Compound Risk	0 pts	15 pts

[Rule 3] New Policy + High Claim	Compound Risk	0 pts	10 pts
Theoretical Score Range (Total)		0 pts	195 pts

Chapter 6. Limitations and Caveats

These criteria have been designed for simulation purposes at the undergraduate academic level. The following differences exist when compared with the underwriting standards of actual insurance companies.

6.1 Arbitrariness of Score Weights and Cutoff Thresholds

The scoring systems of actual insurers are derived from statistical models (logistic regression, GLM, etc.) calibrated against years of loss ratio data. By contrast, the score weights (e.g., "less than 6 months since last claim = 25 points") and the assessment cutoff thresholds (0/26/56/86 points, etc.) in these criteria are hypothetical standards informed by risk prioritization principles from reference documents, and have not been subjected to empirical calibration.

However, the objective of this experiment is not the absolute accuracy of the scoring weights, but rather the comparison of hallucination rates across General AI, RAG AI, and Data Mesh + RAG AI under a uniform standard.

6.2 Absence of Loss Ratio Data

This dataset was originally designed for CRM marketing analysis (identification of renewal offer candidates) and does not include variables necessary for loss ratio calculation, such as actual claim payouts, incident occurrence status, or accident frequency. The Total Claim Amount represents the amount claimed by the policyholder and may differ from the amount actually paid by the insurer.

6.3 Temporal Limitations of Reference Documents

The reference material — a report by the Korea Consumer Agency — is based on data as of October 2007. Accordingly, it may differ from the current underwriting standards, scoring systems, and regulatory environment of contemporary insurance companies.

6.4 Geographic Bias of the Dataset

The dataset covers only five U.S. states (Arizona, California, Nevada, Oregon, and Washington). The average claim amount variance across states is a mere $18 (3.1%), rendering geographic risk differentiation essentially impractical. This differs fundamentally from the loss ratio-based regional criteria employed by domestic Korean insurers.

6.5 Treatment of Prohibited Discrimination Variables

Gender, Education, and Marital Status have been excluded from the underwriting score. Although Employment Status is referenced in the source document as a proxy for occupation, industry, and vehicle use, the Employment Status variable in this dataset reflects employment condition rather than actual vehicle use purpose (e.g., delivery, parcel service), and has therefore not been applied as an underwriting criterion.

6.6 Risk of Circular Logic in the Target Variable

The underwriting results (Gold Standard) generated using these criteria are derived from hypothetical standards, not actual underwriting outcomes. If these results were used to train a machine learning model, the model would learn to replicate the hypothetical criteria rather than actual risk — a circular logic limitation inherent to this design.

Chapter 7. Bias Monitoring Variables

The following variables are not used in the underwriting score calculation. They are retained to examine whether AI assessments disproportionately concentrate additional review or deferred acceptance outcomes among any particular group.

Monitoring Variable	Inspection Content
Gender	Comparison of rates of additional review and acceptance deferral by gender
Education	Review of assessment distribution by education level
Marital Status	Review of assessment distribution by marital status
Income	Review of assessment distribution by income bracket
Employment Status	Review of assessment distribution by employment type (not reflected in underwriting score)

Chapter 8. References

Korea Consumer Agency. (2007). Problems and Improvement Measures in Automobile Insurance Underwriting Criteria. [In Korean]